

TOMTABACKEN, Sweden

WMO: 025490

Lat: 57.4981N

Lon: 14.4645E

Elev: 1168

StdP: 14.09

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	11.9	3.9	7.0	11.4	7.2	8.3	14.6	19.7	32.4	17.9	33.7	4.4	10	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	77.4	61.8	74.3	61.0	71.0	59.6	64.9	72.0	63.1	70.0	61.4	68.1	6.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	88.5	67.2	60.6	82.6	65.5	58.7	77.1	63.9	30.6	72.5	29.2	69.8	27.9	68.3	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	14.0	12.6	DB	5.7	81.5	5.1	2.9	2.1	83.7	-0.9	85.4	-3.8	87.0	-7.4	89.2	(4)
(5)			WB	4.8	67.7	4.9	1.8	1.3	68.9	-1.6	69.9	-4.3	70.9	-7.8	72.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	26.6	27.3	32.4	41.6	50.2	56.5	60.6	58.8	51.9	43.0	35.5	30.2		
	DBStd	13.60	7.40	7.10	7.37	6.63	6.79	5.79	5.68	4.76	4.95	6.35	6.49	7.60		
	HDD50	3571	724	634	545	263	82	7	0	2	35	229	435	615		
	HDD65	8085	1189	1054	1010	701	461	264	164	200	394	683	885	1080		
	CDD50	1006	0	0	0	12	87	201	328	275	92	11	0	0		
	CDD65	44	0	0	0	0	1	8	26	9	0	0	0	0		
	CDH74	303	0	0	0	0	8	78	173	43	1	0	0	0		
	CDH80	29	0	0	0	0	0	4	21	4	0	0	0	0		
(14) Wind	WSAvg	6.6	7.0	6.9	7.0	6.7	6.5	6.1	5.7	5.8	6.4	7.0	7.6	6.9		
(15) Precipitation	PrecAvg	31.8	2.4	1.9	1.7	1.6	2.4	3.2	3.6	3.6	3.0	3.1	2.7	2.6		
	PrecMax	37.0	4.8	4.5	3.1	3.9	5.0	6.3	7.8	7.0	6.5	6.7	4.7	4.9		
	PrecMin	24.7	0.3	0.3	0.6	0.2	0.8	0.3	0.5	0.7	0.4	0.7	1.0	0.5		
	PrecStd	3.7	1.3	0.9	0.7	1.1	1.0	1.5	1.9	1.7	1.7	1.4	1.0	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	45.6	56.1	66.8	75.0	80.0	82.6	78.5	70.2	60.1	50.4	47.1		
		MCWB	42.4	41.2	45.3	52.5	59.3	60.3	63.3	63.6	60.0	54.4	49.5	46.5		
	2%	DB	40.1	41.2	50.4	61.5	71.0	76.2	78.6	74.0	65.8	55.5	47.9	44.4		
		MCWB	39.4	39.4	42.6	49.0	56.4	59.5	63.0	62.4	57.9	53.3	46.8	44.0		
	5%	DB	38.0	38.8	45.6	57.1	67.0	72.0	75.2	70.5	62.4	53.3	46.0	41.9		
		MCWB	37.4	37.3	40.2	46.3	54.4	58.4	61.9	60.8	56.2	51.3	45.1	41.3		
	10%	DB	36.2	36.7	41.7	52.8	62.5	67.9	71.6	67.3	59.5	51.3	44.3	39.7		
		MCWB	35.6	35.4	38.2	44.3	52.5	56.7	60.9	59.7	54.7	49.6	43.5	38.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	42.6	46.9	53.9	61.6	65.5	67.3	67.0	62.2	55.8	50.1	46.7		
		MCDB	43.0	43.9	53.1	64.7	70.8	72.8	75.7	74.1	67.6	57.2	50.4	47.0		
	2%	WB	39.6	39.6	43.3	50.5	58.6	62.0	65.4	64.4	59.2	53.5	47.2	44.2		
		MCDB	39.9	40.5	48.3	58.9	67.0	71.2	73.4	70.0	63.9	55.3	47.7	44.4		
	5%	WB	37.6	37.6	40.9	47.6	56.2	59.5	63.7	62.5	57.2	51.8	45.4	41.6		
		MCDB	37.9	38.7	44.9	54.6	63.0	68.8	71.2	67.6	61.3	53.1	45.9	41.9		
	10%	WB	35.7	35.5	38.8	45.4	53.8	57.6	62.0	60.8	55.4	49.9	43.6	39.2		
		MCDB	36.1	36.5	41.2	51.3	60.6	66.3	69.6	65.8	58.6	51.1	44.2	39.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.5	11.0	15.0	15.8	15.9	14.6	13.1	11.8	8.7	6.4	6.2		
		MCDBR	7.5	9.1	16.8	21.3	21.3	21.9	20.3	17.9	15.6	10.2	7.8	7.8		
		MCWBR	7.7	7.5	10.4	11.9	11.2	10.6	9.0	8.6	8.9	7.6	7.3	7.8		
		MCDBR	7.2	8.2	15.4	18.2	18.2	18.9	17.0	14.9	14.4	9.6	7.2	7.9		
	5% WB	MCWBR	7.5	7.1	10.4	11.1	10.9	10.6	9.0	8.6	9.1	8.0	7.0	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.309	0.326	0.362	0.360	0.350	0.372	0.368	0.339	0.329	0.302	0.264		
	taud		2.291	2.353	2.387	2.358	2.418	2.464	2.415	2.428	2.499	2.455	2.348	2.228		
	Ebn at Noon		190	235	263	270	278	282	272	266	258	226	178	157		
	Edh at Noon		15	22	28	33	33	32	33	31	25	20	14	12		
(44) All-Sky Solar Radiation	RadAvg		117	316	785	1252	1619	1745	1577	1277	871	402	150	83		
	RadStd		22	60	87	171	157	92	202	131	120	72	28	9		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-288	N/A	N/A

Nomenclature: See separate page